Analysis Sequence "V5 analysis all fields"

New image calculated for cytoplasm borders
New image region calculated for mitochondria
SER texture features calculated for mitochondria
Cytoplasm refined

Input Image	Input
maximum projection all planes to adjust for focus use quick tune when possible	**Channel group :** 1 **Sequences :** ALL **Flatfield Correction :** Advanced Brightfield Correction **Stack Processing :** Maximum Projection

Find Nuclei	Input	Method	Output
	Channel : HOECHST 33342 **ROI :** None	**Method :** B Common Threshold : 0.37 Area : > 43 µm² Splitting Coefficient : 5 Individual Threshold : 0.1 Contrast : > 0.2	Output Population : Nuclei

Calculate Morphology Properties	Input	Method	Output
	Population : Nuclei **Region :** Nucleus	**Method :** Standard Area Roundness Perimeter Ratio Width to Length	Property Prefix : Prenucleus

Select Population	Input	Method	Output
	Population : Nuclei	**Method :** Filter by Property Prenucleus Roundness : > 0.5 Prenucleus Ratio Width to Length : > 0.26 Prenucleus Area [µm²] :	Output Population : Nuclei Selected

		< 590.4 Boolean Operations : F1 and F2 and F3	

Select Population (2)	Input	Method	Output
	Population : Nuclei Selected	**Method :** Common Filters Remove Border Objects Region : Nucleus	Output Population : Nuclei Selected Selected

Calculate Morphology Properties (4)	Input	Method	Output
	Population : Nuclei Selected Selected **Region :** Nucleus	**Method :** Standard Area Roundness Perimeter Ratio Width to Length	Property Prefix : Nucleus

Calculate Image	Input	Method	Output
Don't ask, it works somewhat		**Method :** By Formula Formula : 4*A-B+2*C Channel A : MitoTracker Deep Red Channel B : Brightfield Channel C : Alexa 488 Negative Values : Set to Zero Undefined Values : Set to Local Average Output Image Type :	Output Image : Calculated Image

Find Cytoplasm	Input	Method	Output
	Channel : Calculated Image **Nuclei :** Nuclei Selected	**Method :** F Membrane Channel : MitoTracker Deep Red Individual Threshold : 0	

Select Population (3)	Input	Method	Output
remove cells bordering edge	**Population :** Nuclei Selected	**Method :** Common Filters Remove Border Objects Region : Cytoplasm	Output Population : Cytoplasm selected

Calculate	Input	Method	Output

Morphology Properties (3)			
	Population : Cytoplasm selected **Region :** Cell	**Method :** Standard Area Roundness Ratio Width to Length	Property Prefix : Cell

Find Spots	Input	Method	Output
	Channel : Alexa 488 **ROI :** Cytoplasm selected **ROI Region :** Cell	**Method :** B Detection Sensitivity : 0.32 Splitting Sensitivity : 0.811 Calculate Spot Properties	Output Population : Spots

Calculate Intensity Properties	Input	Method	Output
	Channel : MitoTracker Deep Red **Population :** Cytoplasm selected **Region :** Cell	**Method :** Standard Mean	Property Prefix : Intensity Cell MitoTracker Deep Red

Find Image Region (2)	Input	Method	Output
	Channel : MitoTracker Deep Red **ROI :** Cytoplasm selected **ROI Region :** Cell	**Method :** Common Threshold Threshold : 0.4 Split into Objects Area : > 0 px²	Output Population : Total_mt Output Region : Total_mt

Calculate Morphology Properties (5)	Input	Method	Output
	Population : Total_mt **Region :** Total_mt	**Method :** Standard Area	Property Prefix : Total_mt

Calculate Texture Properties	Input	Method	Output
	Channel : MitoTracker Deep Red **Population :** Cytoplasm selected **Region :** Total_mt	**Method :** SER Features Scale : 1 px Normalization by : Kernel SER Spot SER Hole SER Edge SER Ridge	Property Prefix : Total_mt MitoTracker Deep Red

		SER Valley SER Saddle SER Bright SER Dark	

Calculate Intensity Properties (4)	Input	Method	Output
	Channel : MitoTracker Deep Red **Population :** Nuclei Selected **Region :** Cell	**Method :** Standard Mean Median Sum	Property Prefix : Intensity Cell MitoTracker Deep Red

Calculate Intensity Properties (3)	Input	Method	Output
	Channel : Alexa 488 **Population :** Spots **Region :** Spot	**Method :** Standard Mean	Property Prefix : Intensity Spot Alexa 488

Find Image Region	Input	Method	Output
	Channel : Alexa 488 **ROI :** Cytoplasm selected **ROI Region :** Cell	**Method :** Common Threshold Threshold : 0.36 Split into Objects Area : > 0 px²	Output Population : Lipid_droplets Output Region : Lipid_droplets

Calculate Morphology Properties (2)	Input	Method	Output
	Population : Cytoplasm selected **Region :** Lipid_droplets	**Method :** Standard Area	Property Prefix : Lipid_droplets

Calculate Intensity Properties (2)	Input	Method	Output
	Channel : Alexa 488 **Population :** Cytoplasm selected **Region :** Lipid_droplets	**Method :** Standard Mean	Property Prefix : Intensity Lipid_droplets Alexa 488

Define Results	Results
TD: select which	**Method :** List of Outputs **Population : Nuclei**

measurements to export, by mean or median or both etc.	Apply to All : None **Population : Nuclei Selected Selected** Number of Objects Nucleus Area [µm²] : Median Nucleus Roundness : Median Nucleus Ratio Width to Length : Median **Population : Total_mt** Total_mt Area [µm²] : Median **Population : Nuclei Selected** Intensity Cell MitoTracker Deep Red Mean : Median Intensity Cell MitoTracker Deep Red Median : Median Intensity Cell MitoTracker Deep Red Sum : Median **Population : Spots** Number of Objects Relative Spot Intensity : Median Corrected Spot Intensity : Median Spot Area [px²] : ALL **Population : Cytoplasm selected** Cell Area [µm²] : Median Cell Roundness : Median Cell Ratio Width to Length : Median Number of Spots : Median Number of Spots per Area of Cell : Median Intensity Cell MitoTracker Deep Red Mean : ALL Total_mt MitoTracker Deep Red SER Spot 1 px : Median Total_mt MitoTracker Deep Red SER Hole 1 px : Median Total_mt MitoTracker Deep Red SER Edge 1 px : Median Total_mt MitoTracker Deep Red SER Ridge 1 px : Median Total_mt MitoTracker Deep Red SER Valley 1 px : Median Total_mt MitoTracker Deep Red SER Saddle 1 px : Median Total_mt MitoTracker Deep Red SER Bright 1 px : Median Total_mt MitoTracker Deep Red SER Dark 1 px : Median Lipid_droplets Area [µm²] : Median Intensity Lipid_droplets Alexa 488 Mean : Median **Object Results** Population : Nuclei : None Population : Nuclei Selected Selected : None Population : Total_mt : None Population : Nuclei Selected : None Population : Spots : None Population : Lipid_droplets : None Population : Cytoplasm selected : None

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